

Vyacheslav Shen

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ABOUT

I'm a Machine Learning Engineer with over 4 years of experience spanning both industry and academia. My work has primarily focused on medical imaging, and I'm a published author at MICCAI. I'm now seeking a challenging position in ML/AI to broaden my expertise and develop high-load, scalable applications.

EXPERIENCE

- **Nace.AI** , Machine Learning Engineer Dec 2025 - curr.
Palo Alto based AI startup developing audit agents Palo Alto, USA
 - Curated in-house dataset collection for visual document grounding to improve spatial reasoning in Large Vision-Language Models.
- **Cerebra.ai Ltd** , Machine Learning Engineer Nov 2021 - Feb 2023, Nov 2024-Nov 2025
A StartX-accelerated, Central Asia's best MedTech startup Almaty, Kazakhstan
 - Developed an unsupervised anomaly detection system using Generative Adversarial Networks (GANs) with unlabeled CT images of healthy subjects.
 - Conducted research on synthetic data augmentation with Diffusion models, improving segmentation model performance by 3% Dice score.
 - Benchmarking vision-language models for automated stroke detection.
- **Brain Reverse Engineering and Imaging Lab** , AI Researcher Feb 2023 - Jan 2025
A laboratory in KAIST led by Prof. Kim Daeshik, focusing on Deep Learning Daejeon, South Korea
 - Conducted fMRI analysis of light stimulation effects as a non-invasive treatment of Alzheimer's disease using Granger Causality. Improved the accuracy of ViT (+13.1%) & ConvNext (+2.6%) for Alzheimer's classification in a low-data regime using Self-Supervised Learning. Improved fMRI-to-text generation with Contrastive Learning and BLIP2 Vision-Language Model with 8B parameters (+3.6% CLIP-L Similarity & +7.9% METEOR than previous SOTA)

EDUCATION

- **KAIST**, M.S. in Electrical Engineering Mar 2023 - Feb 2025
South Korea's Top 1 research university Daejeon, South Korea
 - Completed with GPA of **4.00/4.3**
- **KAIST**, B.S. in Aerospace and Electrical Engineering Sep 2018 - Feb 2023
South Korea's Top 1 research university Daejeon, South Korea
 - Awarded a **full scholarship** and stipend by the university

PUBLICATIONS C=CONFERENCE, T=THESIS

- [C] **Vyacheslav Shen**, Kassymzhomart Kunanbayev, Donggon Jang, Kim Daeshik. (2025). **Interpretable fMRI Captioning via Contrastive Learning**. 2025 International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI).
- [T] **Vyacheslav Shen**, Kim Daeshik. (2024). **Brain Decoding with Large Language-Vision models**. Master's Thesis.
- [C] Kassymzhomart Kunanbayev, **Vyacheslav Shen**, Kim Daeshik. (2024). **Training ViT with Limited Data for Alzheimer's Disease Classification: an Empirical Study**. In 2024 MICCAI

PROJECTS

- **Chat with Video: RAG project** Sep 2024
Tools: Streamlit, OpenAI API, Docker, Apache Airflow, Elastic Search 
 - Developed real-time and interactive chat functionality using LLM APIs and Streamlit, allowing users to query the video-specific knowledge base. Integrated Apache Airflow for automated evaluation tasks.
 - Implemented and evaluated multiple retrieval methods, achieving up to 84.6% hit rate and 61.7% MRR for content retrieval with Hybrid Search from Elastic Search.

CERTIFICATIONS

- **Coursera**: [Stanford Machine Learning & Django](#) Mar 2021 & Aug 2022
- **DataTalks.Club**: [MLOps Zoomcamp](#) & [LLM Zoomcamp](#) Sep 2022 & Oct 2024
- **Yandex**: [LLM Scaling Week](#) Nov 2025

SKILLS

- **Tech Stack**: Python, Pytorch, Lightning, Clearml, Monai, huggingface, vLLM, Docker, FastAPI

ADDITIONAL INFORMATION

- **Programming & ML**: [Kaggle Expert](#) & [Top 4.5%](#) in [Leetcode](#) contests
- **Teaching**: Lecturer at [outpeer.kz](#)